

AI Code Reviewer Benchmark Brief

Round One - six models, one frozen real codebase, blind independent referee

9-10 September 2026 · CR-BENCH-01 · Executive decision brief

BOTTOM LINE

GLM 5.3 won raw review quality at 76.0 / B. The bigger operational discovery was **GLM 5.3 Flash at 73.2 / B-** - only 2.8 points behind while showing **+3.2 percentage points** of observed session usage versus **+26.5** for GLM 5.3.

Decision: validate GLM 5.3 Flash for routine BIM/FFM review, GLM 5.3 for deeper phase review, and reserve Astra-class reasoning for strategic or high-risk gates.

76.0

RAW QUALITY WINNER

GLM 5.3 - strongest overall raw reviewer

73.2

OPERATIONAL SURPRISE

GLM 5.3 Flash - only 2.8 points behind

+3.2 pp

FLASH SESSION DELTA

vs +26.5 pp for full GLM 5.3

Final leaderboard

BLIND ASTRA REFEREE

#	MODEL	SCORE	GRADE	BLIND ROLE
1	GLM 5.3	76.0	B	Broad senior / phase reviewer
2	GLM 5.3 Flash	73.2	B-	Broad senior / phase reviewer
3	DeepSeek V4 Pro	55.0	C-	Tactical portal/auth reviewer
4	DeepSeek V4 Flash	54.6	D	Security-focused triage
5	MiniMax 3	45.4	D	Tactical migration/module
6	Kimi K2.7 Code	40.8	F	Exploratory UI/integration

What we actually tested

This was a **natural senior-review benchmark**, not a checklist-following contest. Every model reviewed the same dead snapshot under the same raw prompt. No Stark reviewer playbook, expected finding list, scoring rubric, or Astra reference findings were supplied.

THE SPECIMEN

Pinned commit **466083f...**; snapshot independently verified; **357-file SHA-256 manifest**.

THE TRACK

Same target, same raw prompt, same review-only boundary, **no source edits and no coaching**.

THE REFEREE

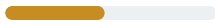
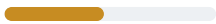
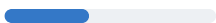
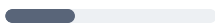
GPT-6 Astra, High reasoning. Reports randomized A-F; identities and economics withheld.

THE CLOSEOUT

Jarvis audited Astra while the mapping remained sealed and accepted the result without score changes.

The 100-point referee instrument

12 DIMENSIONS

Finding validity		15	Important coverage		15
Evidence quality		10	Severity calibration		7
Confidence discipline		7	Security reasoning		8
State / concurrency		8	Architecture / root cause		8
Test-quality reasoning		6	Signal / noise		5
Operational usefulness		6	Independent judgment		5

Why this matters

Finding count alone could not win. False positives, important misses, weak evidence, inflated severity and noisy duplication all cost points.

The blind race

IDENTITY SEPARATED FROM JUDGMENT

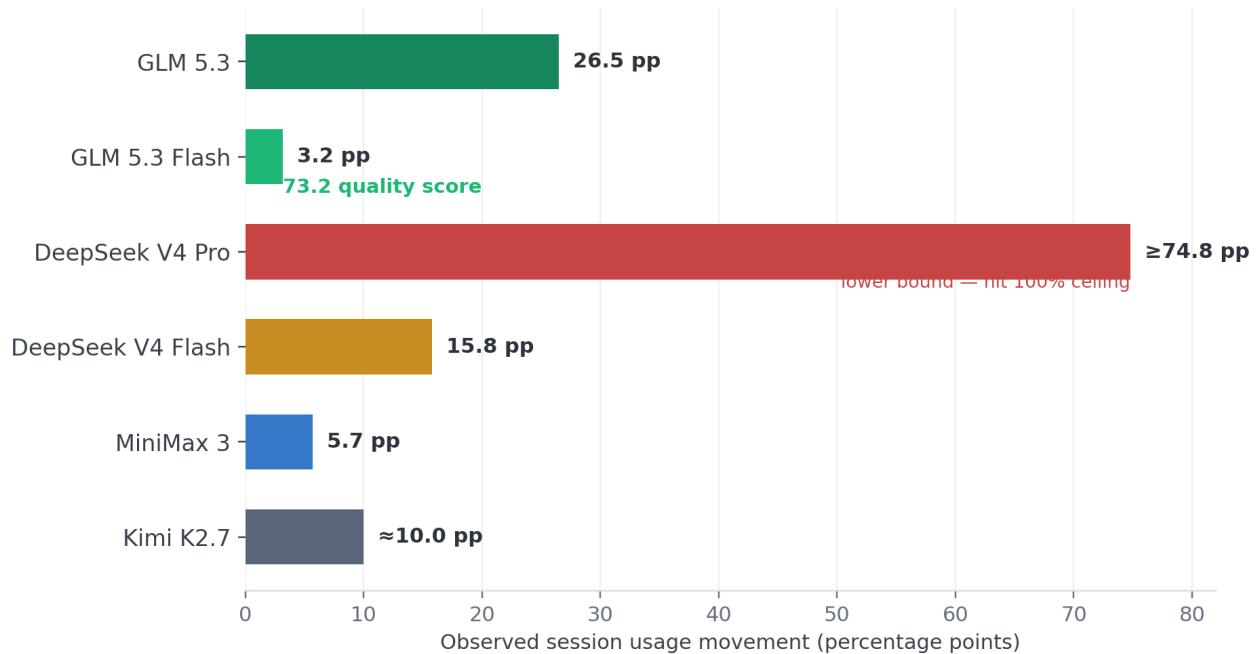
The six reports were randomized to REVIEWER_A through REVIEWER_F. Astra scored those identities first. Only after Astra's report was frozen and Jarvis accepted it was the mapping opened and economics reconnected.

The usage bar chart that changes the story

Observed Ollama session usage

Percentage-point movement during each reviewer run - lower is operationally lighter

Ollama session usage: the cost surprise



GLM 5.3 Flash: the economic shock

73.2 quality at only +3.2 observed session percentage points. In this run it delivered roughly 96% of GLM 5.3's score while showing roughly 12% of GLM 5.3's session-meter movement.

DeepSeek V4 Pro: the opposite signal

55.0 quality, ≥74.8 pp observed session demand, one HTTP 429, two quota windows, and ~1.5 hours of observed recovery. More apparent compute did not buy better review here.

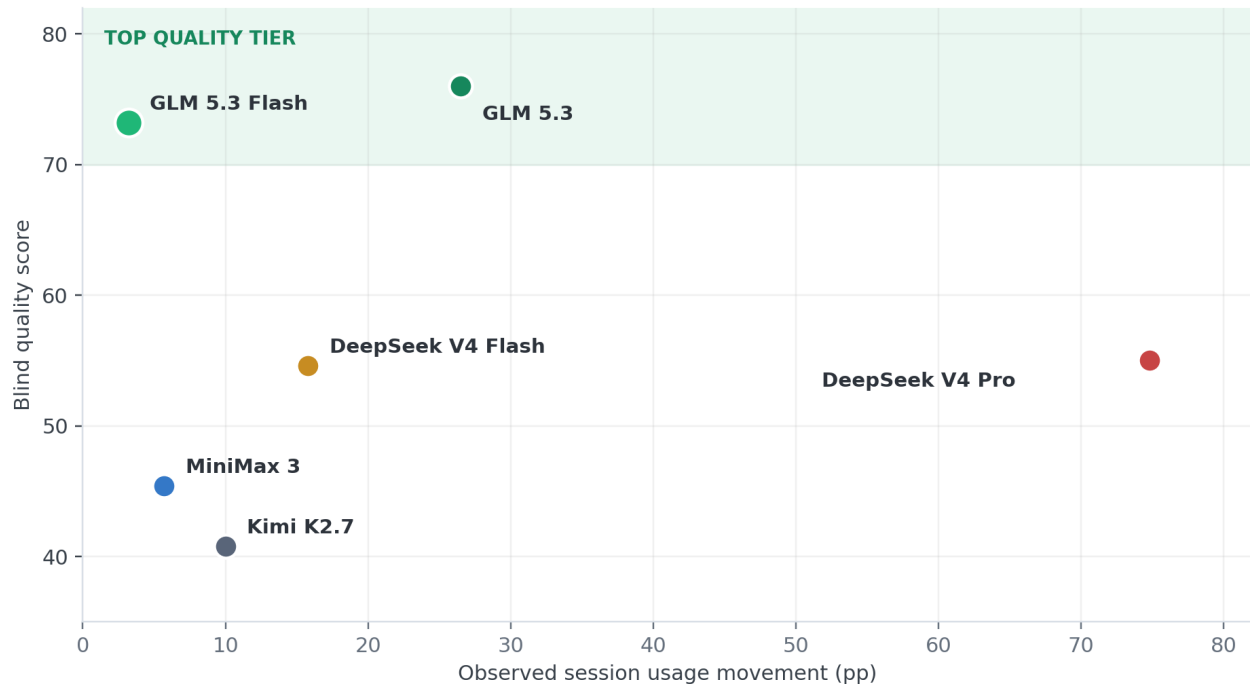
Provider percentage-point meters are operational evidence, not tokens, dollars, normalized compute or benchmark scores. DeepSeek Pro's ≥74.8 pp is a lower bound because its first segment hit the 100% ceiling.

Quality vs operational usage

Who bought the most quality for the meter movement?

Upper-left is the attractive zone: higher quality, lighter observed session usage

Quality vs operational usage



GLM top tier

GLM 5.3 and GLM Flash are the only contestants above 70. Astra called their **2.8-point gap close** enough that a one-point validity reassessment could reverse them.

DeepSeek near-tie

DeepSeek Pro 55.0 vs Flash 54.6 is effectively a **0.4-point tie**. The C-/D boundary makes the difference look larger than it is.

THE CEILING QUESTION

76 is not a ceiling. Round One measured raw, uncoached behavior. The next scientific question is how much a frozen Stark Code Reviewer Playbook can lift the same models.

Six models, six review personalities

GLM 5.3

76.0 / B

WORKED

Best balance of coverage, root cause, tests and verification judgment.

HELD IT BACK

Some secondary claims unverified; missed the precise live Mission Control failure path.

GLM 5.3 Flash

73.2 / B-

WORKED

Strong cross-layer/state reasoning; two fully valid unique contributions; excellent value signal.

HELD IT BACK

Incorrect cookie/SSR advice and unsafe role-remediation assumptions.

DeepSeek V4 Pro

55.0 / C-

WORKED

Concise and relatively precise inside its chosen portal/auth scope.

HELD IT BACK

Missed major agent/instruction trust boundaries; enormous operational demand.

DeepSeek V4 Flash

54.6 / D

WORKED

Found consequential security surfaces and useful triage findings.

HELD IT BACK

Severity inflation, duplicate roots, weak claim control and an invalid redirect mechanism.

MiniMax 3

45.4 / D

WORKED

Useful migration, fixture, archive and concurrency observations.

HELD IT BACK

Too much low-impact cleanup/speculation; important trust-boundary misses.

Kimi K2.7 Code

40.8 / F

WORKED

Useful shared-profile issue and build/test controls.

HELD IT BACK

Invalid/unresolved security claims; critical caller-authorization root missed.

The thing model reputation did not predict

The provider's most operationally demanding contestant did not win. The lightest observed reviewer nearly did. **That is exactly why the benchmark was blinded and why economics were reconnected only after technical scoring.**

What the reviewers were supposed to protect

120 SUBSTANTIVE BLOCKS	73 VALID	29 PARTIAL	4 INVALID	10 DUPLICATE
L01	CRITICAL	Privileged actions omit caller authorization		
L02	HIGH	Target-user restrictions are UI-only		
L03	HIGH	Agent run/history lack identity and ownership binding		
L04	HIGH	Instruction GET/PUT lack authorization		
L05	MIXED	Manifest, UI, mocks and tests disagree		
L06	MIXED	Account creation / trigger contract mismatch		

THE SCORING LESSON

Root-cause compression beat symptom volume. A smaller, precise report could outperform a larger speculative report. The benchmark rewarded evidence and judgment, not verbosity.

Evidence discipline

Severity and evidence were scored as separate axes. Source evidence could support a serious concern without proving deployed exploitability. Conversely, reproducible behavior could still be low severity. That separation helped penalize reviewers who turned every concern into a crisis.

Why we trust the result

PINNED SPECIMEN

Commit 466083f... independently verified against a fresh checkout.

TARGET INTEGRITY

357 / 357 SHA-256 entries remained valid through final closeout.

EQUAL TRACK

Same frozen target and same substantive raw prompt for all six contestants.

RAW PRESERVATION

Original reports were preserved and hashed before adjudication.

BLIND QUALITY

A-F randomized; identities and economics withheld from Astra.

REVEAL DISCIPLINE

Mapping opened only after Jarvis blind adjudication was committed.

Referee limitations we are carrying forward

RF-001 · Blinding isolation

A broad pre-prompt file listing exposed identity-bearing contestant directory names. The mapping and original reports were not read; no material contamination was demonstrated.

RF-002 · SOURCE VERIFICATION

Astra was not authorized to inspect the target source. It followed protocol, but independent finding-level verification was limited.

vNext fix

Physically isolate the referee workspace and consider read-only access to an isolated frozen target. The benchmark instrument and the referee harness are themselves systems that must be tested.

What this round does not prove

One repository and one controlled raw-review round do not establish universal GLM superiority, normalized dollar cost, statistical significance, or deployed exploitability for every finding.

Factory decision

ROUTINE BIM / FFM

GLM 5.3 Flash

73.2 raw quality +
strongest observed
efficiency signal.

PHASE COMPLETION

GLM 5.3

Highest raw score +
broad senior/systems
behavior.

STRATEGIC / HIGH-RISK

GPT-6 Astra

Principal challenge /
adjudication where
expected value justifies
cost.

Proposed routine flow

ENGINEER



REGRESSION



GLM FLASH



REMEDiate



QA / GATE Q

ESCALATE DEEPER REVIEW WHEN CHANGES TOUCH

Auth/authz · RLS/tenant boundaries · payments · DB migrations · external APIs · agent/tool authority
· persistence · concurrency · major state/refactor/security-sensitive work.

Final decision

Quality winner: GLM 5.3 - 76.0 / B.

Operational discovery: GLM 5.3 Flash - 73.2 / B- at +3.2 observed session percentage points.

Factory direction: cheap independent review frequently; deeper review at phase boundaries; principal reasoning at strategic risk.

Next: CR-BENCH-02 championship field; CR-BENCH-03 Stark Playbook Round.

Evidence beats assertion. Reproducibility beats confidence.